



ML for Domain Analytics

Introduction to Machine Learning

Definition

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from historical data and make predictions or decisions without being explicitly programmed for every scenario.

Instead of following fixed rules, ML identifies relationships in data and uses them to predict future outcomes.

Example

Traditional Programming

Employee Data + Rules → Output

Machine Learning

Employee Data + Historical Outcomes → ML Model
→ Prediction

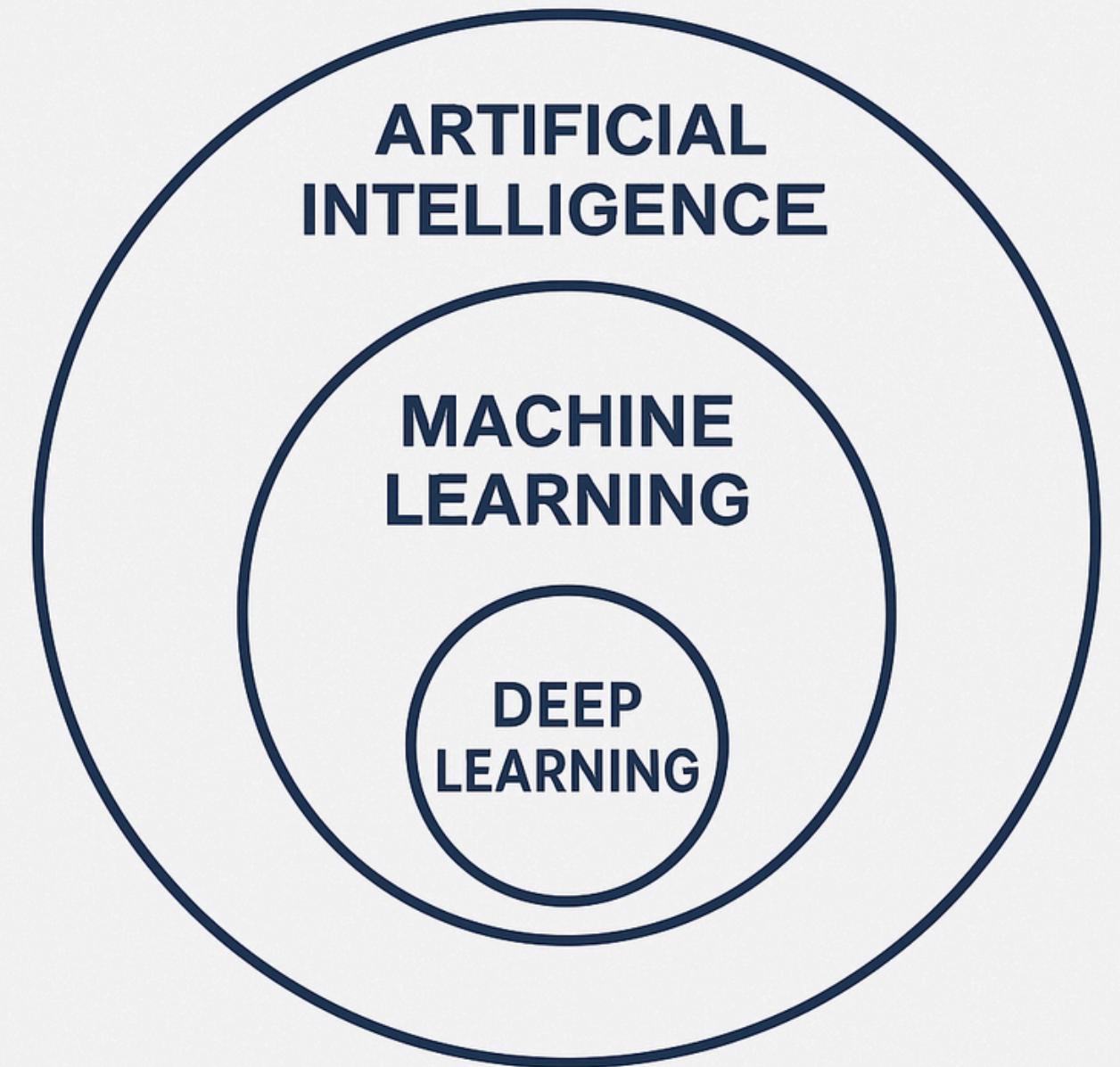
Instead of manually identifying employees who might resign, an ML model learns from historical employee data and predicts future attrition.

Artificial Intelligence vs Machine Learning vs Deep Learning

Artificial Intelligence	Machine Learning	Deep Learning
Mimics human intelligence	Learns from data	Uses neural networks
Broad concept	Subset of AI	Subset of ML
Rule + Learning	Data-driven	Large datasets

Example

- **AI:** ChatGPT
- **Machine Learning:** Employee Attrition Prediction
- **Deep Learning:** Face Recognition



Types of Machine Learning

Type	Data	Goal
Supervised Learning	Labelled	Predict outcome
Unsupervised Learning	Unlabelled	Find hidden groups
Reinforcement Learning	Feedback	Learn by reward

Supervised Learning

Definition

The model learns from labelled historical data where the correct answer is already known.

Input

Customer Details



Output

Creditworthy or Not

The algorithm learns the relationship between customer characteristics and creditworthiness.

Examples

- Credit Prediction
- Credit scoring Prediction
- Fraud Detection
- Financial forecasting

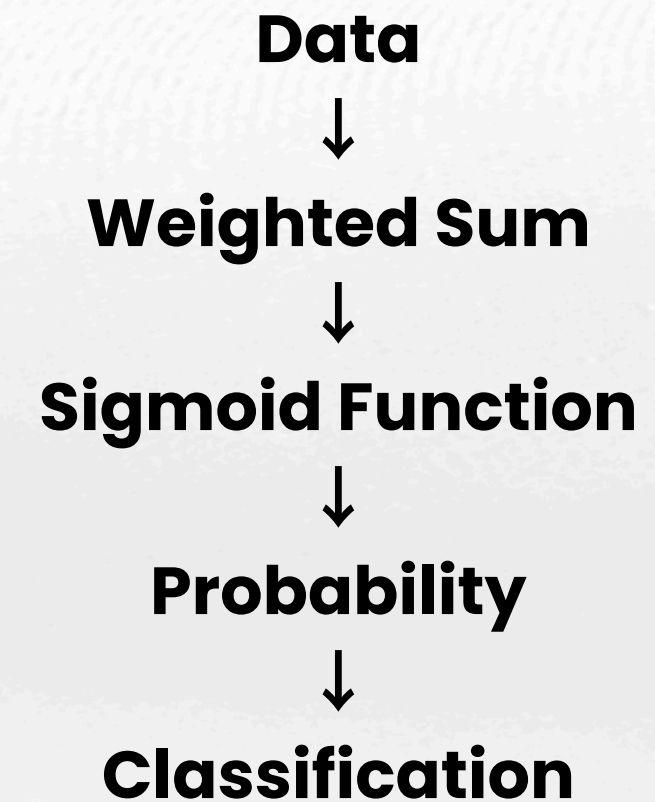
Introduction to Logistic Regression

What is Logistic Regression?

Logistic Regression is a supervised machine learning algorithm used for binary classification problems.

Instead of predicting an exact numerical value, it predicts the probability that an observation belongs to a particular class.

How Logistic Regression Works



Example

Probability = 0.89



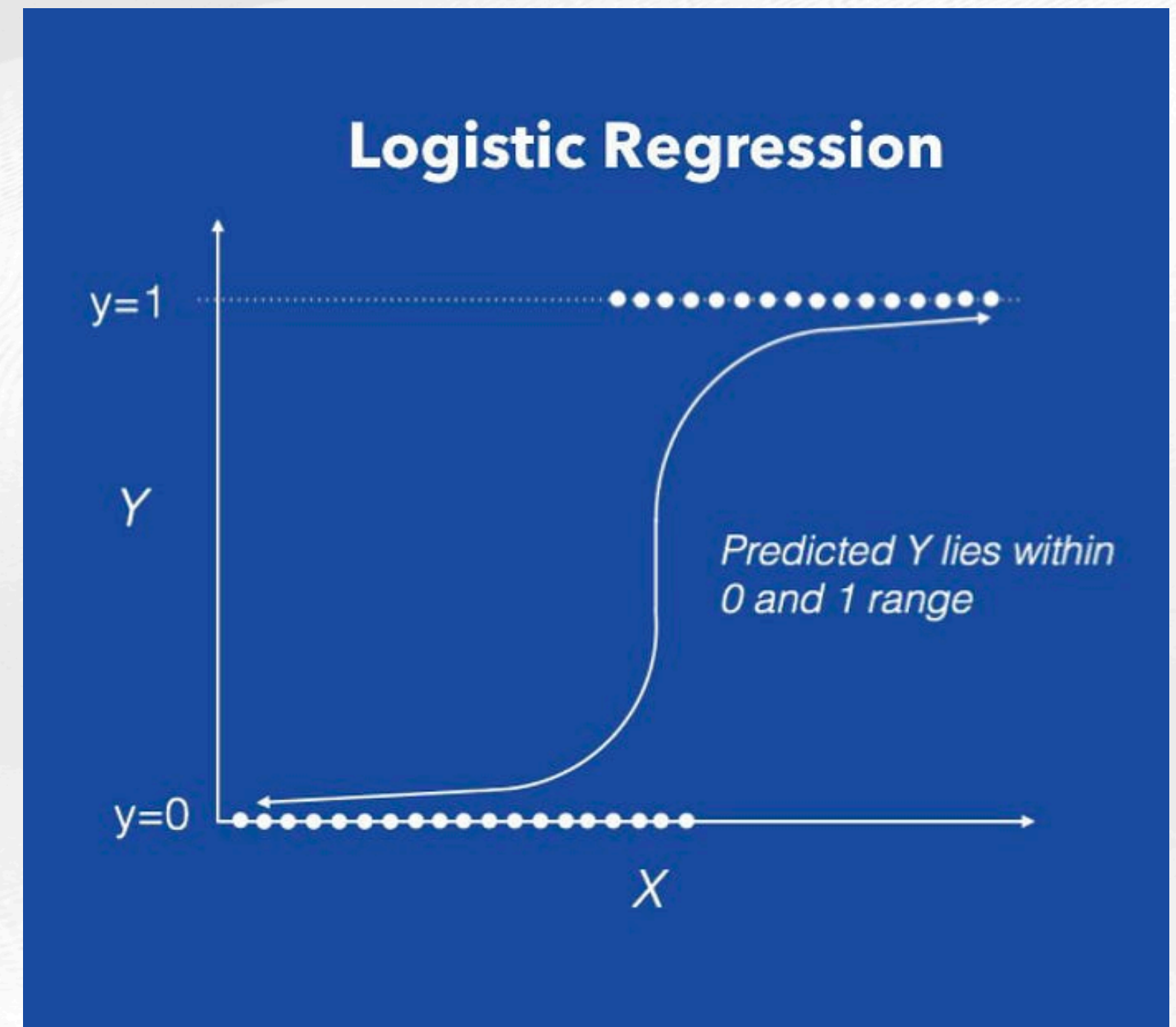
Prediction = Credit Worthy

Sigmoid Function

The Sigmoid Function converts any real number into a value between 0 and 1, making it suitable for probability estimation.

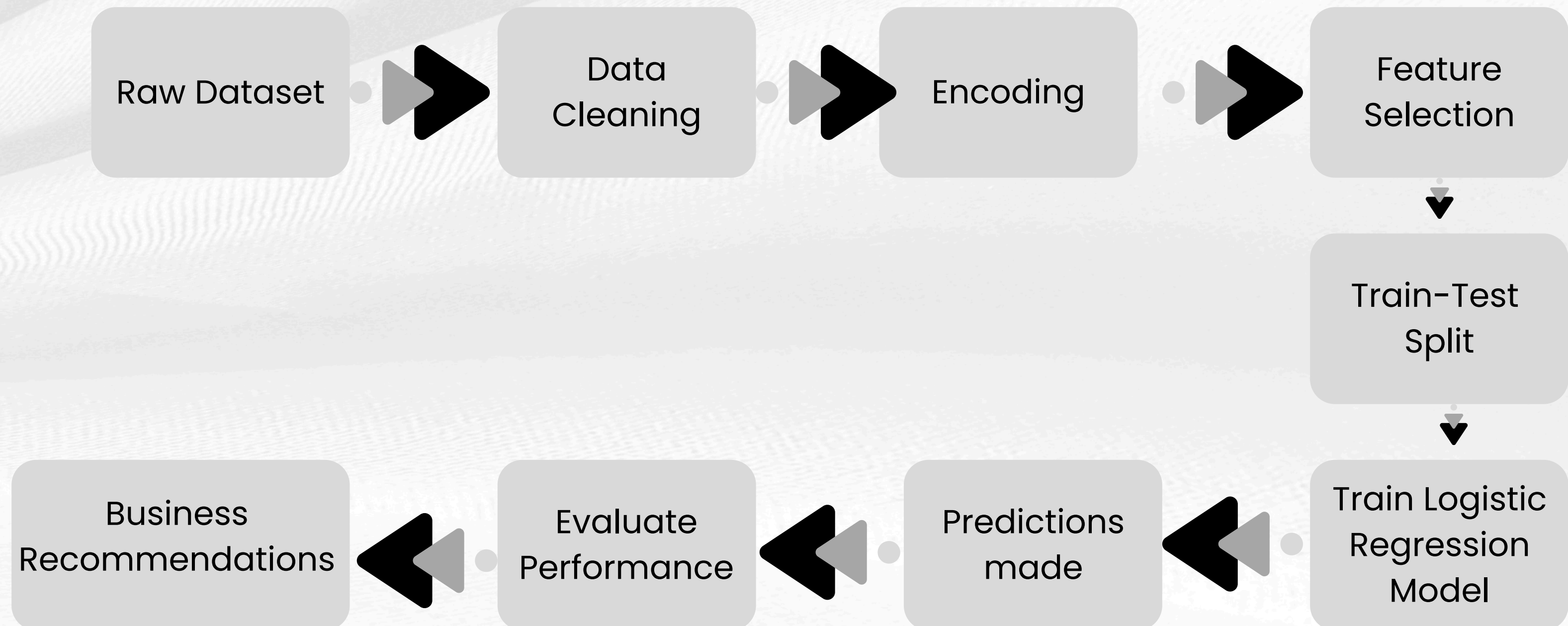
Properties:

- Output always lies between 0 and 1.
- Smooth S-shaped curve.
- Values close to 1 indicate higher likelihood of the positive class.



Building the Prediction Model using Logistic Regression

Machine Learning Pipeline



Training vs Testing Data

Why don't we use the entire dataset for training?

If the model is trained and tested on the same data, it may simply memorize the dataset instead of learning general patterns.

Using separate training and testing datasets helps evaluate how well the model performs on unseen employees.

Dataset	Percentage	Purpose
Training Data	80%	Learn patterns
Testing Data	20%	Evaluate performance

Analogy

Think of preparing for an exam:

- Training Data = Practice questions
- Testing Data = Final exam

A good model should perform well on the final exam, not just the practice questions.

Accuracy and Confusion Matrix

What is a Confusion Matrix?

A Confusion Matrix compares the model's predictions with the actual outcomes.

It helps us understand

- Correct predictions
- Incorrect predictions
- Types of mistakes made by the model

	Predicted CW	Predicted Not CW
Actual CW	True Negative (TN)	False Positive (FP)
Actual Not CW	False Negative (FN)	True Positive (TP)

- **True Positive (TP):** Model correctly predicts the positive class.
- **True Negative (TN):** Model correctly predicts the negative class.
- **False Positive (FP):** Model predicts positive, but the actual class is negative.
- **False Negative (FN):** Model predicts negative, but the actual class is positive.

Accuracy and Confusion Matrix

What is Accuracy?

Definition

Accuracy measures how often the model makes correct predictions.

It answers the question:

"Out of all employees, how many did the model classify correctly?"

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{All predictions}}$$

Example

Suppose we tested the model on 100 customers.

Model Results

- Correctly predicted 88 customer CW
- Incorrectly predicted 12 customers NOT CW

Therefore,

Accuracy

$$88 / 100 = 88\%$$

Interpretation

The model correctly predicts customer creditworthiness (CW) for 88 out of every 100 customers.

Classification Evaluation

Precision

Meaning

Of all the cases predicted as positive, how many were actually positive?

Example

The model identifies 100 customer as CW.

Only 80 actually CW.

Precision=80%

Important When

False positives are costly.

Recall

Meaning

Of all actual positive cases, how many did the model correctly identify?

Example

100 people actually CW.

The model identifies 90.

Recall=90%

Important When

Missing a positive case is costly.

F1-Score

F1-Score is the harmonic mean of Precision and Recall.

Classification Evaluation

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall}$$